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Github Link: <https://github.com/192424318-pradeep-1980/CSA0920-Java>

CO 1 – ASSESSEMENT 1

Question:

HashMap vs Hashtable

Write a Java program that creates both a HashMap and a Hashtable, and attempts to insert a null key or a null value into each. Display the outcome of each attempt (success or exception/error) to demonstrate that HashMap permits null keys/values while Hashtable does not.

Concepts: HashMap, Hashtable, null handling.

Code:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        HashMap<String, String> hm = new HashMap<>();
        Hashtable<String, String> ht = new Hashtable<>();

        hm.put(null, "Java");
        hm.put("A", null);
```

```

System.out.println("HashMap: " + hm);

try {
    ht.put(null, "Java");
} catch (Exception e) {
    System.out.println("Hashtable null key: " + e);
}

try {
    ht.put("A", null);
} catch (Exception e) {
    System.out.println("Hashtable null value: " + e);
}
}
}

```

Output:

```

HashMap : {null=Java, 1=null}
Hashtable null key: java.lang.NullPointerException:
Cannot invoke "Object.hashCode()" because "key" is null
Hashtable null value: java.lang.NullPointerException

```

Result :

- HashMap successfully inserts a null key and a null value.
- Hashtable throws a NullPointerException when a null key or a null value is inserted.
- Therefore, HashMap allows null keys and values, whereas Hashtable does not.